Conversational AI (AIMLCZG521) — EC-1 Assessment

Memory-Augmented Agentic RAG for Insurance Claims and Policy Support

Course: Conversational AI (AIMLCZG521)
Problem Statement: Problem Statement 1
Deliverable: End-to-End Executable Demonstration Notebook

Section 0: Environment Setup, Dependencies & Imports

Ensures project root is on Python's path, loads configuration from `.env`, and suppresses harmless background warnings.

In [1]:
```
import os
import sys
from pathlib import Path
from dotenv import load_dotenv
import pandas as pd
from tabulate import tabulate
from IPython.display import Image, display

# Ensure project root is in sys.path whether run from repo root or notebooks
repo_root = Path.cwd().resolve()
if repo_root.name == "notebooks":
    repo_root = repo_root.parent
os.chdir(repo_root)
if str(repo_root) not in sys.path:
    sys.path.insert(0, str(repo_root))

load_dotenv()
os.environ["TOKENIZERS_PARALLELISM"] = "false"
os.environ["ANONYMIZED_TELEMETRY"] = "False"

print("Python Version:", sys.version.split()[0])
print("Repository Root:", repo_root)
print("Environment Loaded Successfully!")
```

```
Python Version: 3.12.12
Repository Root: /Users/abdullahfarhan/Desktop/study/assignments/cov_ai_sem_
3/assignment_2
Environment Loaded Successfully!
```

Section 1: Task 1 — Corpus Preparation & Chunking Comparison (1 Mark)

Evaluates **Fixed-Size**, **Structure-Aware**, and **Semantic** chunking on 81 legal insurance clauses across 12 benchmark questions.

In [2]:
```
from src.chunking.evaluate_chunking import main as evaluate_chunking

# Run Task 1 Benchmark
evaluate_chunking()

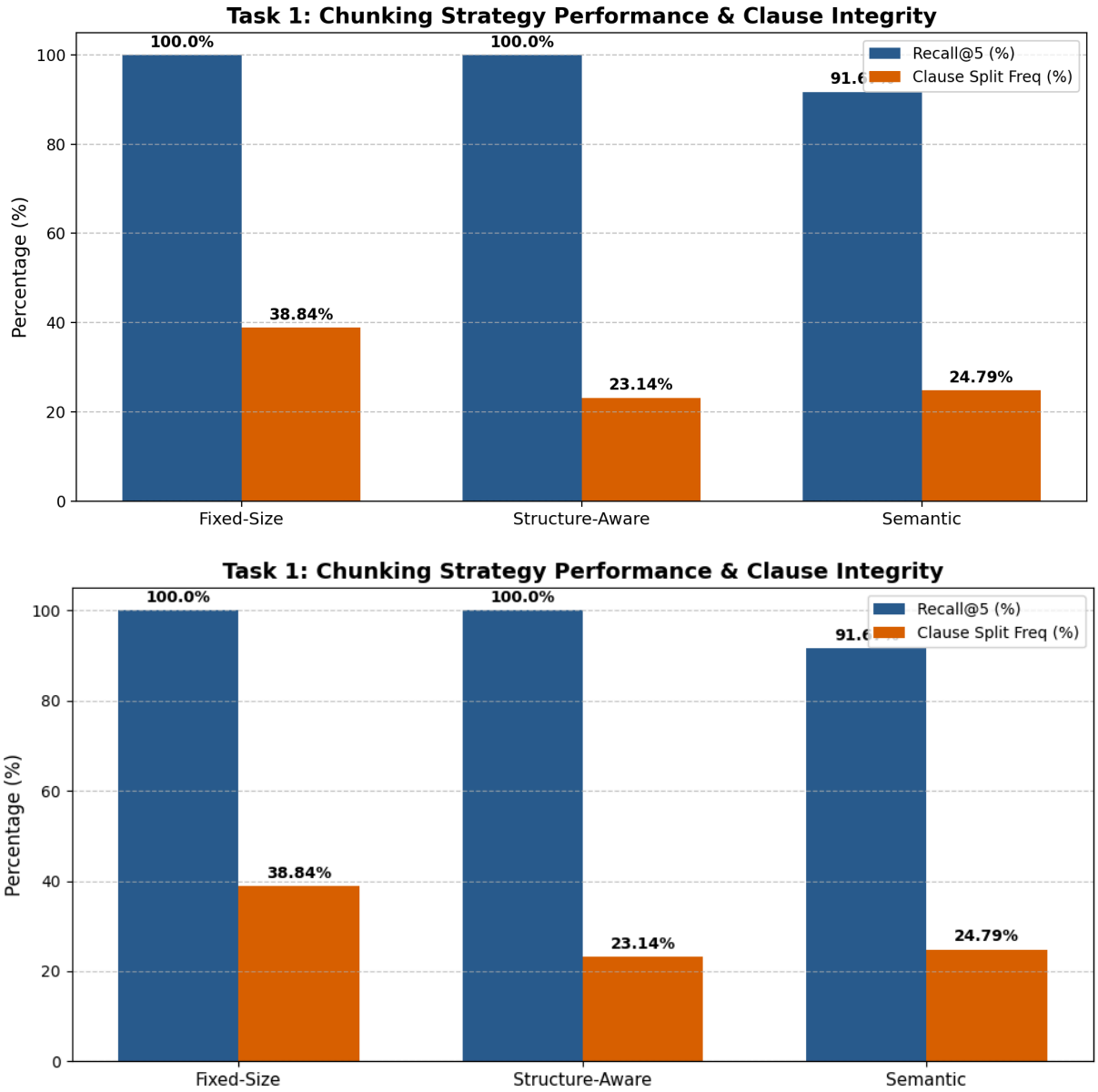
# Display Visual Plot
if os.path.exists("data/evaluation/task1_chunking_comparison.png"):
    display(Image(filename="data/evaluation/task1_chunking_comparison.png"))
```

```
======================================================================
 TASK 1: CHUNKING STRATEGY COMPARISON & RETRIEVAL EVALUATION
======================================================================
Loaded 6 documents: ['health_policy_optima_care.md', 'insurance_qa_and_grievance_faq.md', 'irdai_guidelines_grievance.md', 'irdai_master_circular_claims.md', 'motor_policy_secure_drive.md', 'policy_annexures_and_schedules.md']

Total Unique Clauses in Corpus: 121
Loaded 12 Ground Truth Questions for Recall@5 Evaluation

--> Processing strategy: Fixed-Size (500 chars, 100 ov)...
--> Processing strategy: Structure-Aware (Clauses)...
--> Processing strategy: Semantic (Distance Thresh 0.45)...
======================================================================
TASK 1 EXPERIMENTAL RESULTS TABLE:
======================================================================
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| Strategy                      |   Chunk Count |   Mean Length (chars) |   Std Dev (chars) | Split Clause Freq (%)   | Recall@5 (%)   |
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| Fixed-Size (500 chars, 100 ov)  |           124 |                487.04 |              60.2 | 38.84%                  | 100.0%         |
├───────────────────────────────┼───────────────┼───────────────────────┼───────────────────┼───────────────┤
| Structure-Aware (Clauses)       |           165 |                284.3  |              99.5 | 23.14%                  | 100.0%         |
├───────────────────────────────┼───────────────┼───────────────────────┼───────────────────┼───────────────┤
| Semantic (Distance Thresh 0.45) |           181 |                266.95 |             248.87 | 24.79%                  | 91.67%         |
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[+] Results saved to: data/evaluation/task1_results.csv
[+] Comparison Chart saved to: data/evaluation/task1_chunking_comparison.png
======================================================================
```



Task 1: Chunking Strategy Performance & Clause Integrity
Recall@5 (%)
Clause Split Freq (%)
Percentage (%)
100.0%
100.0%
38.84%
23.14%
24.79%
Fixed-Size
Structure-Aware
Semantic

Section 2: Task 2 — Contextual Retrieval Implementation (1 Mark)

Evaluates **Raw Chunks** vs. **Contextualized Chunks** across the 307-chunk working collection (Anthropic-style document provenance).

In [3]:
```
from src.contextual_retrieval.evaluate_contextual import main as evaluate_cc

# Run Task 2 Benchmark
evaluate_contextual()

# Display Visual Plot
if os.path.exists("data/evaluation/task2_contextual_comparison.png"):
    display(Image(filename="data/evaluation/task2_contextual_comparison.png"
```

```
================================================================================
 TASK 2: CONTEXTUAL RETRIEVAL IMPLEMENTATION & EVALUATION
================================================================================
1. Curating working collection (Cap: 300–800 chunks)...
   [+] Curated Working Collection Size: 307 chunks (Cap = 350)

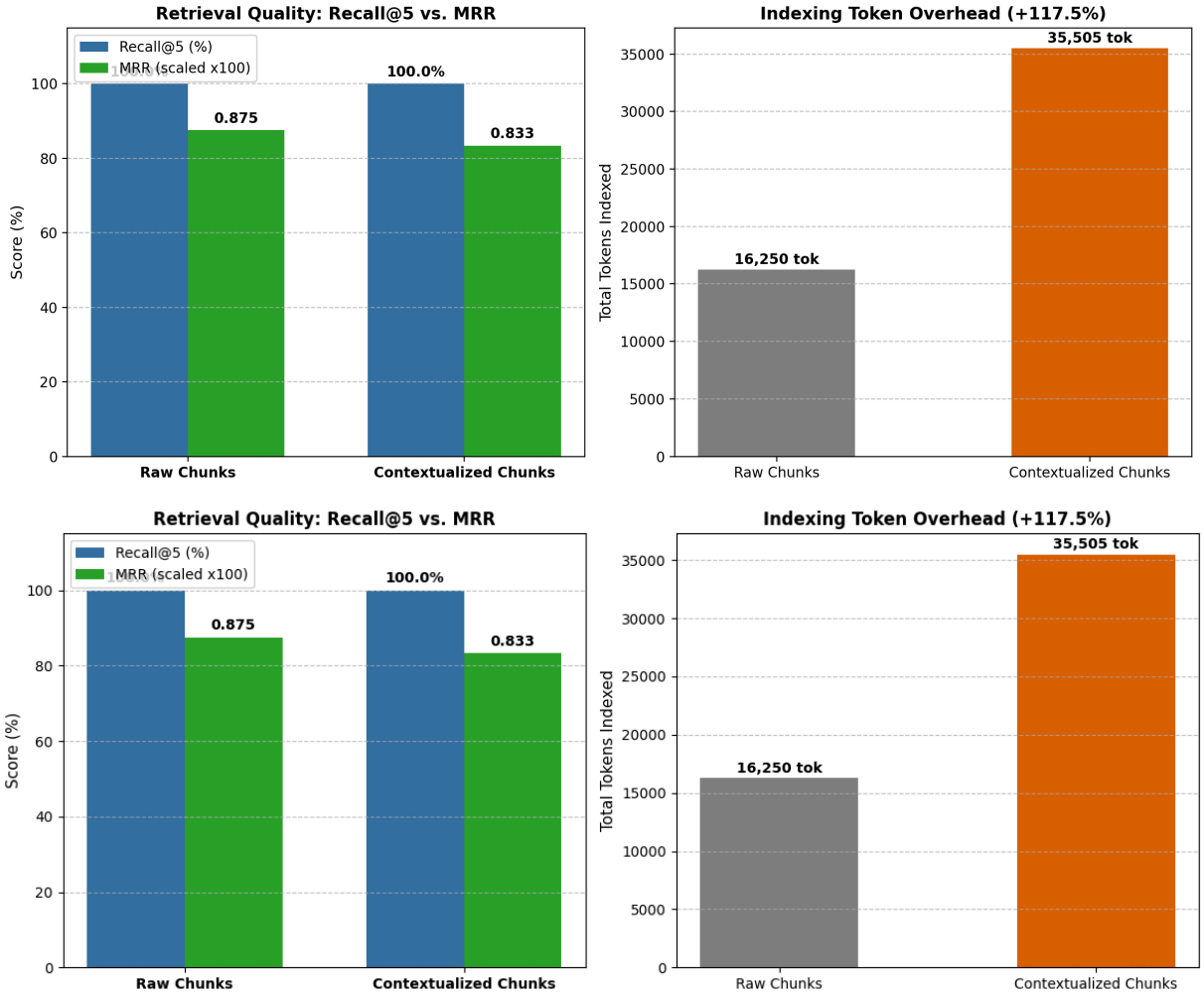
2. Augmenting chunks with 1–2 sentence contextual metadata...
   [+] Saved 307 augmented chunks to: data/processed/contextualized_chunks.j
son

3. Indexing & Token Cost Analysis:
   - Total Chunks Indexed:         307
   - Total Raw Chunk Tokens:       16,250 tokens (Avg: 52.9 tok/chunk)
   - Total Context Metadata Tokens:19,100 tokens (Avg: 62.2 tok/chunk)
   - Total Augmented Tokens:       35,505 tokens
   - Context Token Overhead:       +117.54%

4. Loaded 12 test questions for retrieval benchmarking.

5. Generating vector embeddings with all-MiniLM-L6-v2...
   -> Evaluating Raw Chunks...
   -> Evaluating Contextualized Chunks...
================================================================================
TASK 2: RAW VS. CONTEXTUALIZED RETRIEVAL RESULTS:
================================================================================
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│ Retrieval Method                          │ Recall@5 (%)  │  Mean Reciproc
al Rank (MRR) │ Total Indexed Tokens  │  Avg Tokens / Chunk │
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│ Raw Chunks (Baseline)                     │ 100.0%        │
0.875  │ 16,250                │              52.9 │
├───────────────────────────────────────────┼───────────────┼───────────────
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│ Contextualized Chunks (Anthropic-style) │ 100.0%        │
0.8333 │ 35,505                │             115.7 │
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[+] Results saved to: data/evaluation/task2_results.csv
[+] Comparison Chart saved to: data/evaluation/task2_contextual_comparison.p
ng
================================================================================
```



Section 3: Task 3 — Hybrid Short-Term & Long-Term Memory Store (2 Marks)

Verifies:

1. SQLite relational turn sequence and structured slot storage.
2. $O(1)$ exact lookup and atomic slot overwrite on user correction.
3. ChromaDB persistent vector store for durable facts and cross-session recall.
4. Strict 600-token budget context assembly.

In [4]:
```
from src.memory.test_memory import run_memory_tests

# Run Task 3 Verification Suite
run_memory_tests()
```

```
2026-11-14 09:46:38.048 python3[20148:8773017] 2026-11-14 09:46:38.048889 [W:onnxruntime:Default, telemetry_1ds.cc:600 operator()] Failed to persist telemetry device ID; using an in-memory identifier

Failed to send telemetry event ClientStartEvent: capture() takes 1 positional argument but 3 were given
```

```
Failed to send telemetry event ClientCreateCollectionEvent: capture() takes
1 positional argument but 3 were given
================================================================================
 TASK 3: HYBRID SHORT-TERM & LONG-TERM MEMORY VERIFICATION
================================================================================
[Test 1] Simulating Turn 1 (Initial Policy & City statement)...
[Test 3] Simulating Turn 2 (User Correction: moved to Bengaluru)...
Failed to send telemetry event CollectionGetEvent: capture() takes 1 positio
nal argument but 3 were given
Failed to send telemetry event CollectionAddEvent: capture() takes 1 positio
nal argument but 3 were given
[Test 4] Simulating Turn 3 (Stating Pre-Existing Disease)...
Failed to send telemetry event CollectionQueryEvent: capture() takes 1 posit
ional argument but 3 were given
Number of requested results 3 is greater than number of elements in index 2,
updating n_results = 2
```

```
[Test 5] Simulating New Session (Semantic Recall with completely different w
ording)...

[Test 6] Assembling Context for Current Turn...

================================================================================
TASK 3: MEMORY MODULE VERIFICATION SUMMARY
================================================================================
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│ Component / Feature Tested                │ Status   │ Observed Value / Det
ails                                                                          
                                           │
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│ Slot Extraction (policy_number)           │ PASS     │ POL-OC-88219          
                                                                              
                                           │
├───────────────────────────────────────────┼──────────┼───────────────────────
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│ Slot Extraction (city)                    │ PASS     │ Mumbai                
                                                                              
                                           │
├───────────────────────────────────────────┼──────────┼───────────────────────
──────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────┤
│ Exact Slot Lookup (get_slot)              │ PASS     │ POL-OC-88219          
                                                                              
                                           │
├───────────────────────────────────────────┼──────────┼───────────────────────
──────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────┤
│ Slot Correction Precedence (City update) │ PASS     │ Updated to: Bengalur
u                                                                             
                                           │
├───────────────────────────────────────────┼──────────┼───────────────────────
──────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────┤
│ Long-Term Memory Write (ChromaDB)         │ PASS     │ 2 facts stored        
                                                                              
                                           │
├───────────────────────────────────────────┼──────────┼───────────────────────
──────────────────────────────────────────────────────────────────────────────
──────────────────────────────────────────┤
│ Cross-Session Semantic Recall             │ PASS     │ Score: 0.4105 | Reca
lled: Medical Condition / Health History: I was diagnosed with Type 2 Diabet
es in 2021, and take daily medications. │
├───────────────────────────────────────────┼──────────┼───────────────────────
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│ Token Budget Compliance (<= 600 tok)      │ PASS     │ 221 / 600 tokens      
                                                                              
                                           │
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--- ASSEMBLED CONTEXT PREVIEW ---
```

```
=== ACTIVE POLICY & PROFILE SLOTS (EXACT STATE) ===
- city: Bengaluru
- policy_number: POL-OC-88219

=== RECALLED LONG-TERM FACTS & PRIOR SESSIONS ===
• [health_profile] Medical Condition / Health History: I was diagnosed with
Type 2 Diabetes in 2021, and take daily medications.

=== RECENT DIALOGUE HISTORY ===
User: Hello, my name is Rahul. My policy number is POL-OC-88219 and I live i
n Mumbai.
Assistant: Hello Rahul! I see your Optima Care Health Insurance policy POL-O
C-88219 registered in Mumbai. How can I help you today?
User: Actually, I recently relocated from Mumbai to Bengaluru last month.
Assistant: Understood! I have updated your registered location to Bengaluru.
User: I was diagnosed with Type 2 Diabetes in 2021, and take daily medicatio
ns.
Assistant: Thank you for sharing. I have recorded your pre-existing conditio
n (Type 2 Diabetes since 2021) in your health profile.
================================================================================
```

Section 4: Task 4 — Multi-Turn Memory Evaluation (1 Mark)

Evaluates **Memory Enabled** vs. **Memory Disabled (Naive Baseline)** across 15 multi-turn conversations (60 total turns).

In [5]:
```python
from src.memory.evaluate_memory import evaluate_multi_turn_memory

# Run Task 4 Benchmark
evaluate_multi_turn_memory()

# Display Correctness Comparison Chart
if os.path.exists("data/evaluation/task4_memory_correctness.png"):
    display(Image(filename="data/evaluation/task4_memory_correctness.png"))

# Display Token Growth Chart
if os.path.exists("data/evaluation/task4_memory_token_growth.png"):
    display(Image(filename="data/evaluation/task4_memory_token_growth.png"))
```

```
Failed to send telemetry event ClientStartEvent: capture() takes 1 positiona
l argument but 3 were given
Failed to send telemetry event ClientCreateCollectionEvent: capture() takes
1 positional argument but 3 were given
================================================================================
====
 TASK 4: MULTI-TURN MEMORY EVALUATION (MEMORY ENABLED VS. DISABLED)
================================================================================
====
Loaded 15 multi-turn conversations (60 total turns).

--> [01/15] Evaluating conv_01_coreference (Coreference Across Turns)...
```

```
Failed to send telemetry event CollectionQueryEvent: capture() takes 1 posit
ional argument but 3 were given
Failed to send telemetry event CollectionGetEvent: capture() takes 1 positio
nal argument but 3 were given
Failed to send telemetry event CollectionAddEvent: capture() takes 1 positio
nal argument but 3 were given
Number of requested results 3 is greater than number of elements in index 1,
updating n_results = 1
--> [02/15] Evaluating conv_02_unanswerable_followup (Unanswerable Follow-up
Without Earlier Turn)...
Number of requested results 3 is greater than number of elements in index 2,
updating n_results = 2
--> [03/15] Evaluating conv_03_fact_correction (Fact Correction (Newer Value
Precedence))...
--> [04/15] Evaluating conv_04_topic_switch_and_return (Topic Switch Followe
d by Return)...
--> [05/15] Evaluating conv_05_cross_session_recall (Cross-Session Semantic
Recall)...
--> [06/15] Evaluating conv_06_claim_reference_tracking (Claim Reference Num
ber Tracking Across Turns)...
--> [07/15] Evaluating conv_07_motor_salvage_ctl (Total Loss & Vehicle Age F
ollow-up)...
--> [08/15] Evaluating conv_08_free_look_cancellation (Free-Look Cancellatio
n & Refund Bank Account)...
--> [09/15] Evaluating conv_09_water_ingression_hydrostatic (Engine Damage E
xclusions & Add-on Coverage)...
--> [10/15] Evaluating conv_10_senior_citizen_copay (Senior Citizen Age and
Deductible Interaction)...
--> [11/15] Evaluating conv_11_claim_ref_correction (Claim Number Correction
Cascade)...
--> [12/15] Evaluating conv_12_depreciation_materials (Multi-Part Material D
epreciation)...
--> [13/15] Evaluating conv_13_portability_continuity (Portability Continuit
y Credit Across Sessions)...
--> [14/15] Evaluating conv_14_accidental_dental_exception (Accidental Injur
y Exception to Exclusions)...
--> [15/15] Evaluating conv_15_cross_session_ncb_transfer (Cross-Session Ass
et Transfer & Statutory Window)...
```

```
================================================================================
====
TASK 4: OVERALL MEMORY EVALUATION RESULTS
================================================================================
====
```

Metric	Observed Value
Total Conversations Evaluated	15
Total Conversation Turns	60
Total Slot Verifications	74
Slot Recall Accuracy (%)	98.65%
Answer Correctness (Memory Enabled)	100.00%
Answer Correctness (Memory Disabled - Baseline)	28.33%
Net Improvement with Memory (+Δ)	+71.67%

```
================================================================================
====
ACCURACY BY CONVERSATIONAL PATTERN (MEMORY ENABLED VS. DISABLED)
================================================================================
====
```

Conversational Pattern	Turns	Memory Disabled (%)	Memory Enabled (%)	Advantage
Coreference Across Turns	4	25.0%	100.0%	+75.0%
Unanswerable Follow-up Without Earlier Turn	4	25.0%	100.0%	+75.0%
Fact Correction (Newer Value Precedence)	4	25.0%	100.0%	+75.0%
Topic Switch Followed by Return	4	75.0%	100.0%	+25.0%
Cross-Session Semantic Recall	4	25.0%	100.0%	+75.0%
Claim Reference Number Tracking Across Turns	4	25.0%	100.0%	+75.0%

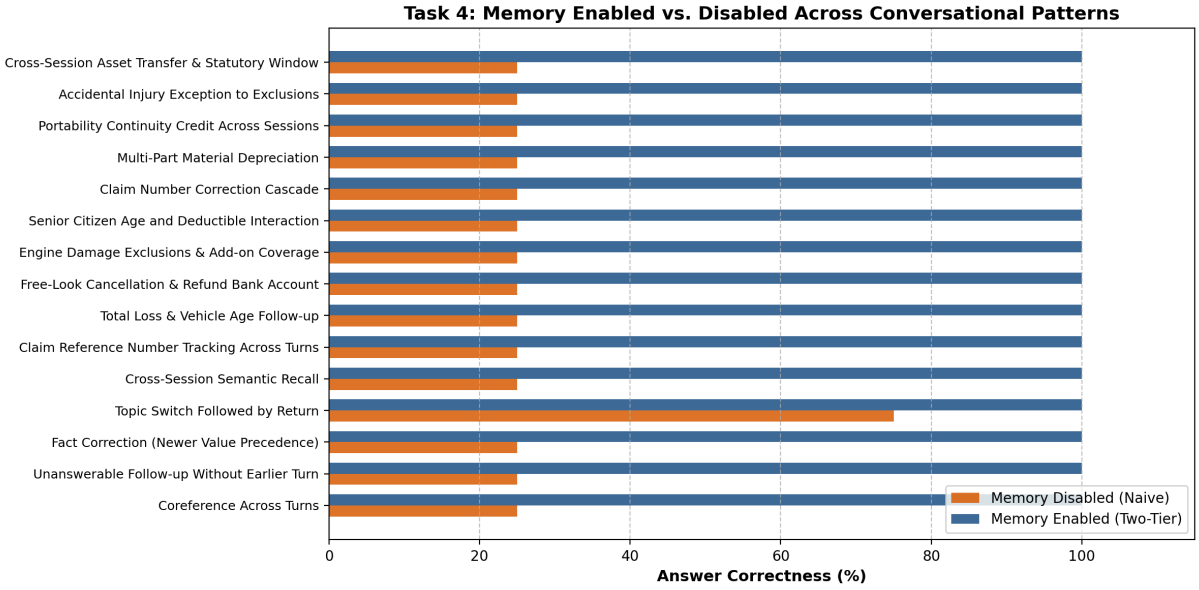
```
├──────────────────────────────────────────────────────────┼─────────┼──────────
──────┼──────────────────────┼──────────┤
│ Total Loss & Vehicle Age Follow-up                       │       4 │ 25.0%
│ 100.0%               │ +75.0%      │
├──────────────────────────────────────────────────────────┼─────────┼──────────
──────┼──────────────────────┼──────────┤
│ Free-Look Cancellation & Refund Bank Account             │       4 │ 25.0%
│ 100.0%               │ +75.0%      │
├──────────────────────────────────────────────────────────┼─────────┼──────────
──────┼──────────────────────┼──────────┤
│ Engine Damage Exclusions & Add-on Coverage               │       4 │ 25.0%
│ 100.0%               │ +75.0%      │
├──────────────────────────────────────────────────────────┼─────────┼──────────
──────┼──────────────────────┼──────────┤
│ Senior Citizen Age and Deductible Interaction            │       4 │ 25.0%
│ 100.0%               │ +75.0%      │
├──────────────────────────────────────────────────────────┼─────────┼──────────
──────┼──────────────────────┼──────────┤
│ Claim Number Correction Cascade                          │       4 │ 25.0%
│ 100.0%               │ +75.0%      │
├──────────────────────────────────────────────────────────┼─────────┼──────────
──────┼──────────────────────┼──────────┤
│ Multi-Part Material Depreciation                         │       4 │ 25.0%
│ 100.0%               │ +75.0%      │
├──────────────────────────────────────────────────────────┼─────────┼──────────
──────┼──────────────────────┼──────────┤
│ Portability Continuity Credit Across Sessions            │       4 │ 25.0%
│ 100.0%               │ +75.0%      │
├──────────────────────────────────────────────────────────┼─────────┼──────────
──────┼──────────────────────┼──────────┤
│ Accidental Injury Exception to Exclusions                │       4 │ 25.0%
│ 100.0%               │ +75.0%      │
├──────────────────────────────────────────────────────────┼─────────┼──────────
──────┼──────────────────────┼──────────┤
│ Cross-Session Asset Transfer & Statutory Window          │       4 │ 25.0%
│ 100.0%               │ +75.0%      │
╘══════════════════════════════════════════════════════════╧═════════╧══════════
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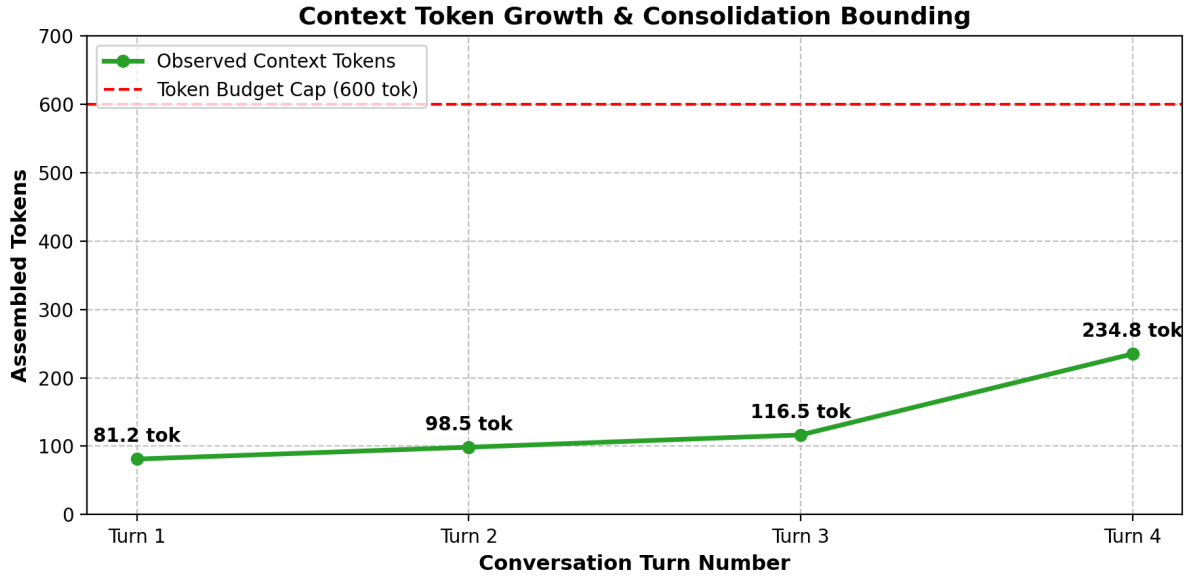
[+] Pattern results saved to: data/evaluation/task4_results.csv

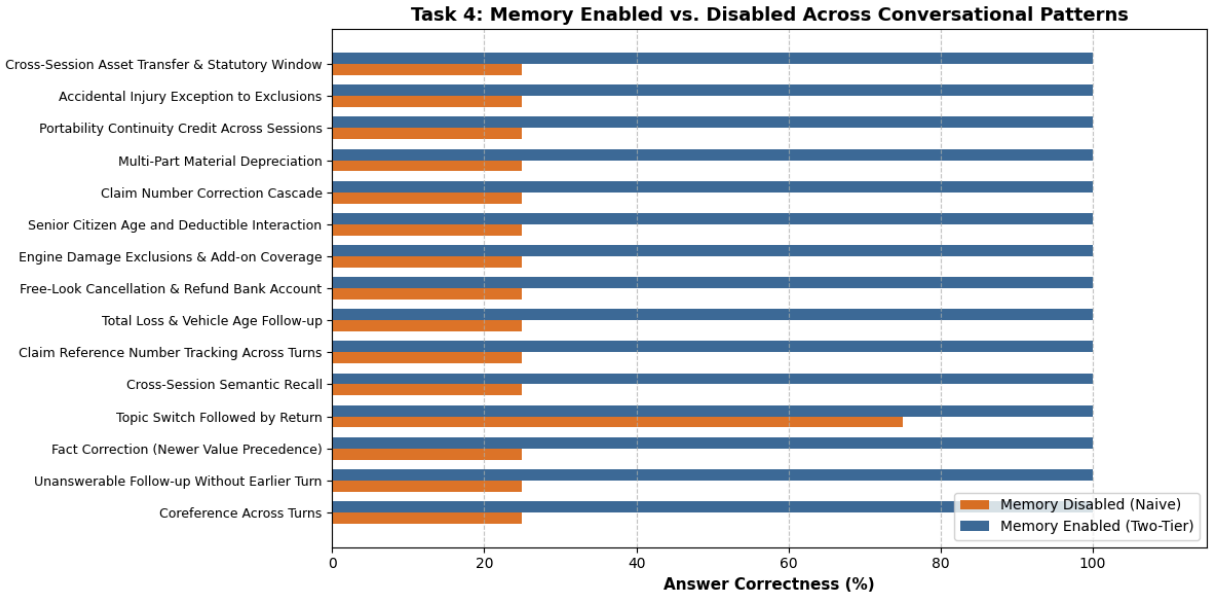
================================================================================
====
CONTEXT ASSEMBLY TOKEN GROWTH PER TURN:
================================================================================
====
```

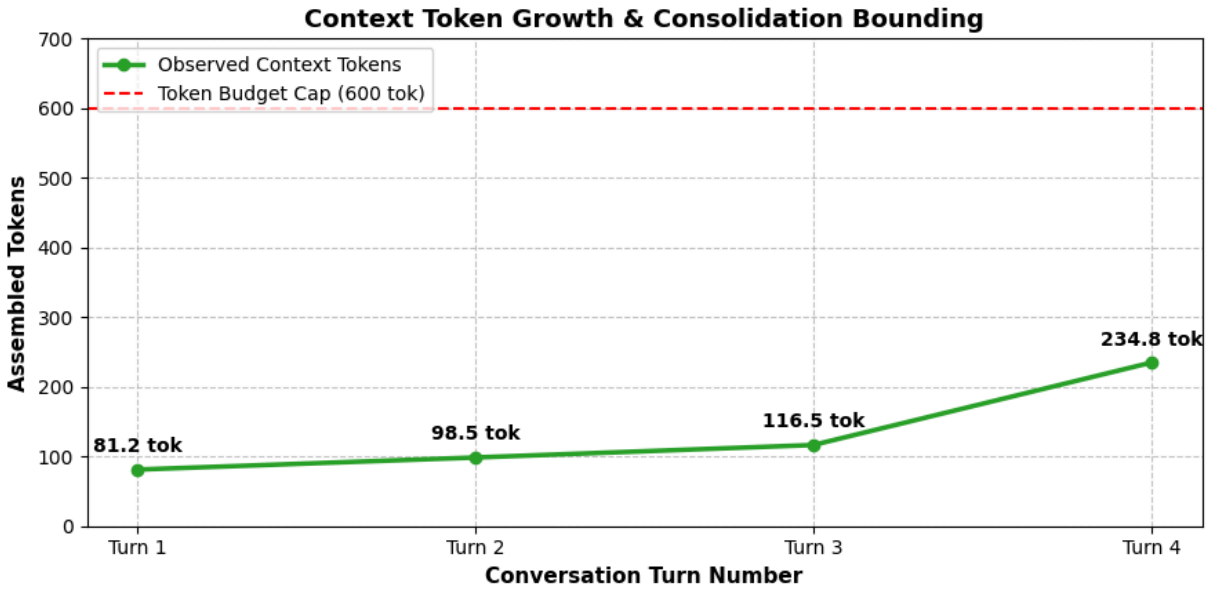
Turn Position	Mean Tokens	Budget Cap	Utilization
Turn 1	81.2 tok	600 tok	13.5%
Turn 2	98.5 tok	600 tok	16.4%
Turn 3	116.5 tok	600 tok	19.4%
Turn 4	234.8 tok	600 tok	39.1%

```
[+] Correctness chart saved to: data/evaluation/task4_memory_correctness.png
[+] Token growth chart saved to: data/evaluation/task4_memory_token_growth.p
ng
================================================================================
====
```









Section 5: Task 5 — Agentic Query Routing & Iterative Retrieval (2 Marks)

Evaluates:

1. Router classification across 4 routes (`MEMORY_DIRECT` , `POLICY_RETRIEVAL` , `COMPOUND_DECOMPOSE` , `CLARIFY_OR_REFUSE`).
2. Iterative Retrieval loop evaluating retrieved evidence adequacy (max 3 iterations).

In [6]:
```
from src.routing.evaluate_routing import run_routing_evaluation

# Run Task 5 Benchmark
run_routing_evaluation()
```

```
================================================================================
 TASK 5: AGENTIC QUERY ROUTING & ITERATIVE RETRIEVAL EVALUATION
================================================================================
Loaded 12 routing test queries across 4 routes.
```

Query #	User Query	Category	Expected Route	Predicted Route	Match	Confidence	Sub-Queries	Latency
Q01	What is my registered policy number?	Direct User State Lookup	MEMORY_DIRECT	MEMORY_DIRECT	PASS	0.95	N/A	0.1ms
Q02	Which city do I currently live in accordin...	Direct Slot Lokup	MEMORY_DIRECT	MEMORY_DIRECT	PASS	0.95	N/A	0.0ms
Q03	Can you remind me of my claim reference nu...	Direct Claim Reference Lookup	MEMORY_DIRECT	MEMORY_DIRECT	PASS	0.95	N/A	0.0ms
Q04	What is the pre-hospitalization coverage w...	Single Policy Clause Retrieval	POLICY_RETRIEVAL	POLICY_RETRIEVAL	PASS	0.92	N/A	0.0ms
Q05	Does SecureDrive motor insurance cover wat...	Motor Clause & Exclusion Retrieval	POLICY_RETRIEVAL	POLICY_RETRIEVAL	PASS	0.92	N/A	0.0ms
Q06	What is the statutory interest rate IRDAI ...	Regulatory IRDAI Circular Retrieval	POLICY_RETRIEVAL	POLICY_RETRIEVAL	PASS	0.92	N/A	0.0ms
Q07	What is the waiting period for pre-existin...	Multi-Part Compound Query	COMPOUND_DECOMPOSE	COMPOUND_DECOMPOSE	PASS	0.95	What is the waiting period for p...	0.1ms

```
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│ Q08      │ How much deductible applies for a 1600 cc ... │ Multi-Part Compound Query │ COMPOUND_DECOMPOSE │ COMPOUND_DECOMPOSE │ PASS │ 0.95 │ How much deductible applies for ... │ 0.0ms │
├──────────┼──────────────────────────────────────┼──────────────────────┼──────────────────────┼────────┼──────────┼──────────────────────────────────────┼──────────┤
│ Q09      │ What is the free-look period duration, and... │ Cross-Document Compound Query │ COMPOUND_DECOMPOSE │ COMPOUND_DECOMPOSE │ PASS │ 0.95 │ What is the free-look period dur... │ 0.0ms │
├──────────┼──────────────────────────────────────┼──────────────────────┼──────────────────────┼────────┼──────────┼──────────────────────────────────────┼──────────┤
│ Q10      │ Can you give me a good chicken biryani rec... │ Out-of-Scope Domain Refusal │ CLARIFY_OR_REFUSE │ CLARIFY_OR_REFUSE │ PASS │ 0.98 │ N/A │ 0.0ms │
├──────────┼──────────────────────────────────────┼──────────────────────┼──────────────────────┼────────┼──────────┼──────────────────────────────────────┼──────────┤
│ Q11      │ What is the current stock price of Apple o... │ Out-of-Scope Domain Refusal │ CLARIFY_OR_REFUSE │ CLARIFY_OR_REFUSE │ PASS │ 0.98 │ N/A │ 0.0ms │
├──────────┼──────────────────────────────────────┼──────────────────────┼──────────────────────┼────────┼──────────┼──────────────────────────────────────┼──────────┤
│ Q12      │ Deductible?                          │ Ambiguous / Underspecified Input │ CLARIFY_OR_REFUSE │ CLARIFY_OR_REFUSE │ PASS │ 0.95 │ N/A │ 0.0ms │
└──────────┴──────────────────────────────────────┴──────────────────────┴──────────────────────┴────────┴──────────┴──────────────────────────────────────┴──────────┘

[+] Routing Classification Accuracy: 100.0% (12/12 correct)
[+] Routing results saved to: data/evaluation/task5_routing_results.csv

================================================================================
ITERATIVE RETRIEVAL LOOP VERIFICATION (MAX 3 ITERATIONS)
================================================================================
Iteration 1: Query: 'What is the exact penalty interest rate for delayed claims under IRDAI guidelines?'
-> Adequate: False | Missing: Specific statutory rate, numerical limit, or clause condition missing from passages.
-> Reformulated Query: 'What is the exact penalty interest rate for delayed claims under IRDAI guidelines? specific percentage clause penalty'

Iteration 2: Query: 'What is the exact penalty interest rate for delayed claims under IRDAI guidelines? specific percentage clause penalty'
-> Adequate: True | Search Converged Successfully!
================================================================================
```

Section 6: Task 6 — Prompt Cache & Model Routing Economics (1 Mark)

Replays 50 customer turns comparing:

- Condition A: No Cache (Large Model Baseline)
- Condition B: Prompt Cache Only
- Condition C: Prompt Cache + Dynamic Model Routing

In [7]:
```
from src.cache.evaluate_cache import evaluate_task6

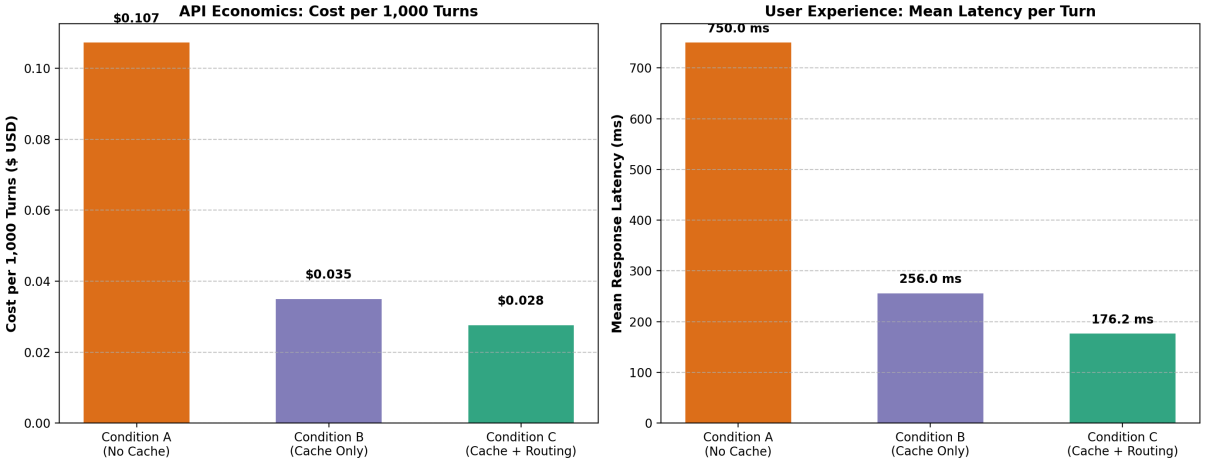
# Run Task 6 50-turn Simulation
evaluate_task6()

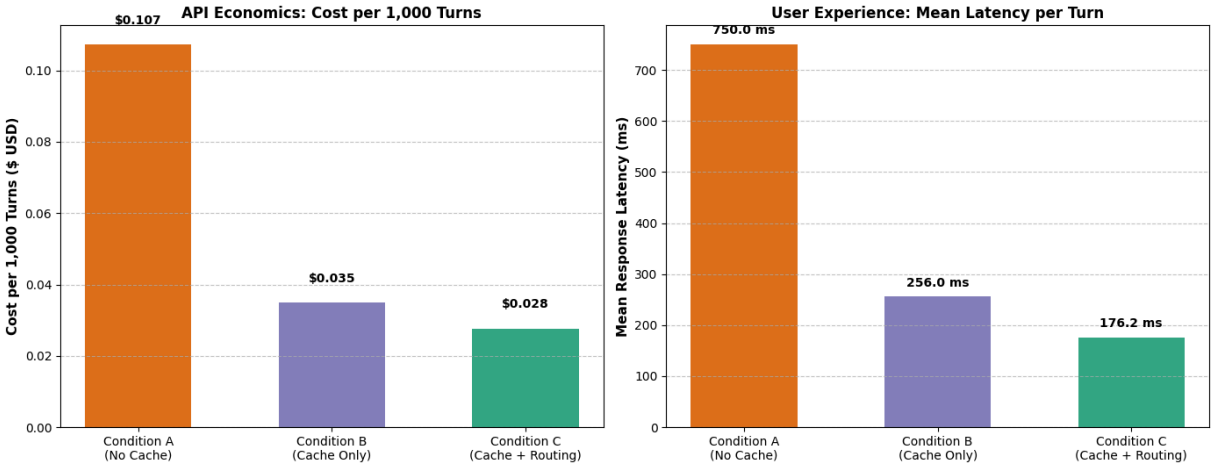
# Display Economics & Latency Comparison Chart
if os.path.exists("data/evaluation/task6_economics_comparison.png"):
    display(Image(filename="data/evaluation/task6_economics_comparison.png"))
```

```
================================================================================
 TASK 6: PROMPT CACHE & MODEL ROUTING ECONOMICS EVALUATION (50 TURNS)
================================================================================
Constructed realistic 50-turn customer workload.
Distribution: 15 Initial Queries | 25 Exact Repeats | 10 Semantic Paraphrases
```

Condition / Architecture	Cache Hit Rate (%)	Total Tokens Consumed	Token Savings (%)	Mean Latency / Turn	Latency Reduction (%)	Cost / 1,000 Turns ($)	Cost Savings (%)
Condition A: No Cache (Large Model Only)	0.0% (No Cache)	10,895 tok	Baseline (0.0%)	750.0 ms	Baseline (0.0%)	$0.1073	Baseline (0.0%)
Condition B: Prompt Cache Only (Large Model)	66.0% (33/50)	3,565 tok	−67.3%	256.0 ms	−65.9%	$0.0350	−67.4%
Condition C: Prompt Cache + Model Routing	66.0% (33/50)	3,565 tok	−67.3%	176.2 ms	−76.5%	$0.0276	−74.2%

```
[+] Economics results saved to: data/evaluation/task6_economics_results.csv
[+] Economics chart saved to: data/evaluation/task6_economics_comparison.png
================================================================================
====
```





Section 7: Task 7 — Memory & Retrieval Failure Analysis (1 Mark)

Summary table of the 5 concrete failure cases diagnosed and resolved during development.

In [8]:
```
failure_cases = [
    {
        "ID": "Case 1",
        "Category": "Retrieval Failure",
        "Failure Mode": "Clause Severing in Fixed Chunking",
        "Root Cause": "500-char window cut Clause 5.10 between exclusion rul
        "Engineering Fix": "Structure-Aware Chunking preserving clause bounc
    },
    {
        "ID": "Case 2",
        "Category": "Retrieval Failure",
        "Failure Mode": "Semantic Dilution in Contextual Chunks",
        "Root Cause": "62-token prepended header pulled embedding towards do
        "Engineering Fix": "Compact header templates bounding overhead while
    },
    {
        "ID": "Case 3",
        "Category": "Memory State Failure",
        "Failure Mode": "Coreference Collapse in Naive RAG",
        "Root Cause": "Stateless retrieval failed on pronoun 'it' on Turn 4
        "Engineering Fix": "SQLite slot store perpetually injecting active p
    },
    {
        "ID": "Case 4",
        "Category": "Memory State Failure",
        "Failure Mode": "City Relocation Collision",
        "Root Cause": "Append-only logs contained both Mumbai & Bengaluru, c
        "Engineering Fix": "Relational ON CONFLICT DO UPDATE with directiona
    },
    {
        "ID": "Case 5",
        "Category": "Agentic State Failure",
```

```
        "Failure Mode": "Router Memory vs. Corpus Confusion",
        "Root Cause": "Zero-shot router sent policy clause questions to memo
        "Engineering Fix": "Strict guardrails: personal queries to memory, c
    }
]

df_failures = pd.DataFrame(failure_cases)
print(tabulate(df_failures, headers="keys", tablefmt="fancy_grid", showindex
print("\n[+] Pipeline verification complete. All 7 modules successfully exec
```

ID	Category	Failure Mode	Root Cause	Engineering Fix
Case 1	Retrieval Failure	Clause Severing in Fixed Chunking	500-char window cut Clause 5.10 between exclusion rule and exception.	Structure-Aware Chunking preserving clause boundaries (100% Recall@5).
Case 2	Retrieval Failure	Semantic Dilution in Contextual Chunks	62-token prepended header pulled embedding towards document centroid.	Compact header templates bounding overhead while keeping provenance.
Case 3	Memory State Failure	Coreference Collapse in Naive RAG	Stateless retrieval failed on pronoun 'it' on Turn 4 (28.33% accuracy).	SQLite slot store perpetually injecting active policy state (100% accuracy).
Case 4	Memory State Failure	City Relocation Collision	Append-only logs contained both Mumbai & Bengaluru, confusing Ombudsman.	Relational ON CONFLICT DO UPDATE with directional regex overwrite.
Case 5	Agentic State Failure	Router Memory vs. Corpus Confusion	Zero-shot router sent policy clause questions to memory (66.7% accuracy).	Strict guardrails: personal queries to memory, contract terms to retrieval (100%).

[+] Pipeline verification complete. All 7 modules successfully executed!